Program 8: Creating an Simple Gradle Project.

Step 1:Create a directory in Devops called MyGradleApp , go to the directory and execute the command gradle init –type java-application

```
sumanth@sumanth-VirtualBox: ~/Devops
sumanth@sumanth-VirtualBox:~$ cd Devops
sumanth@sumanth-VirtualBox:~/Devops$ mkdir MyGradleApp
sumanth@sumanth-VirtualBox:~/Devops$ cd MyGradleApp
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ ls
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ gradle init --type java-application
Starting a Gradle Daemon (subsequent builds will be faster)

BUILD SUCCESSFUL in 12s
2 actionable tasks: 2 executed
```

Step 2: Display Tree using tree command

Step 3: Display The build.gradle in text editor

```
Open    build.gradle
        ~/Devops/MyGradleApp    Save
 * This build file was generated by the Gradle 'init' task.
 *
 * This generated file contains a sample Java project to get you started.
 * For more details take a look at the Java Quickstart chapter in the Gradle
 * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial_java_projects.html
 */

// Apply the java plugin to add support for Java
apply plugin: 'java'

// Apply the application plugin to add support for building an application
apply plugin: 'application'

// In this section you declare where to find the dependencies of your project
repositories {
    // Use jcenter for resolving your dependencies.
    // You can declare any Maven/Ivy/file repository here.
    jcenter()
}

dependencies {
    // This dependency is found on compile classpath of this component and consumers.
    compile 'com.google.guava:guava:23.0'

    // Use JUnit test framework
    testCompile 'junit:junit:4.12'
}

// Define the main class for the application
mainClassName = 'App'


task hello{
doLast{
println'good morning'
}
}
```

Step 3: Now build using the command gradle build

```
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ gradle build
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.pom
Download https://jcenter.bintray.com/com/google/guava/guava-parent/23.0/guava-parent-23.0.pom
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-parent/1.14/animal-sniffer-parent-1.14.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/mojo-parent/34/mojo-parent-34.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_parent/2.0.18/error_prone_parent-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/codehaus-parent/4/codehaus-parent-4.pom
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.pom
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.jar
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.jar
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.jar
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.jar
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.jar

BUILD SUCCESSFUL in 13s
7 actionable tasks: 7 executed
```

Step 4: Display the New Tree Structure

```
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ tree
.
├── build
│   ├── classes
│   │   └── java
│   │       ├── main
│   │       │   └── App.class
│   │       └── test
│   │           └── AppTest.class
│   ├── distributions
│   │   ├── MyGradleApp.tar
│   │   └── MyGradleApp.zip
│   ├── libs
│   │   └── MyGradleApp.jar
│   ├── reports
│   │   └── tests
│   │       └── test
│   │           ├── classes
│   │           │   └── AppTest.html
│   │           ├── css
│   │           │   ├── base-style.css
│   │           │   └── style.css
│   │           ├── index.html
│   │           ├── js
│   │           │   └── report.js
│   │           └── packages
│   │               └── default-package.html
│   ├── scripts
│   │   ├── MyGradleApp
│   │   └── MyGradleApp.bat
│   ├── test-results
│   │   └── test
│   │       ├── binary
│   │       │   ├── output.bin
│   │       │   ├── output.bin.idx
│   │       │   └── results.bin
│   │       └── TEST-AppTest.xml
```

```
│   │       └── TEST-AppTest.xml
│   │   └── tmp
│   │       ├── compileJava
│   │       ├── compileTestJava
│   │       └── jar
│   │           └── MANIFEST.MF
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── settings.gradle
└── src
    ├── main
    │   └── java
    │       └── App.java
    └── test
        └── java
            └── AppTest.java

29 directories, 26 files
```

Step 5: Execute gradle run to execute the gradle task

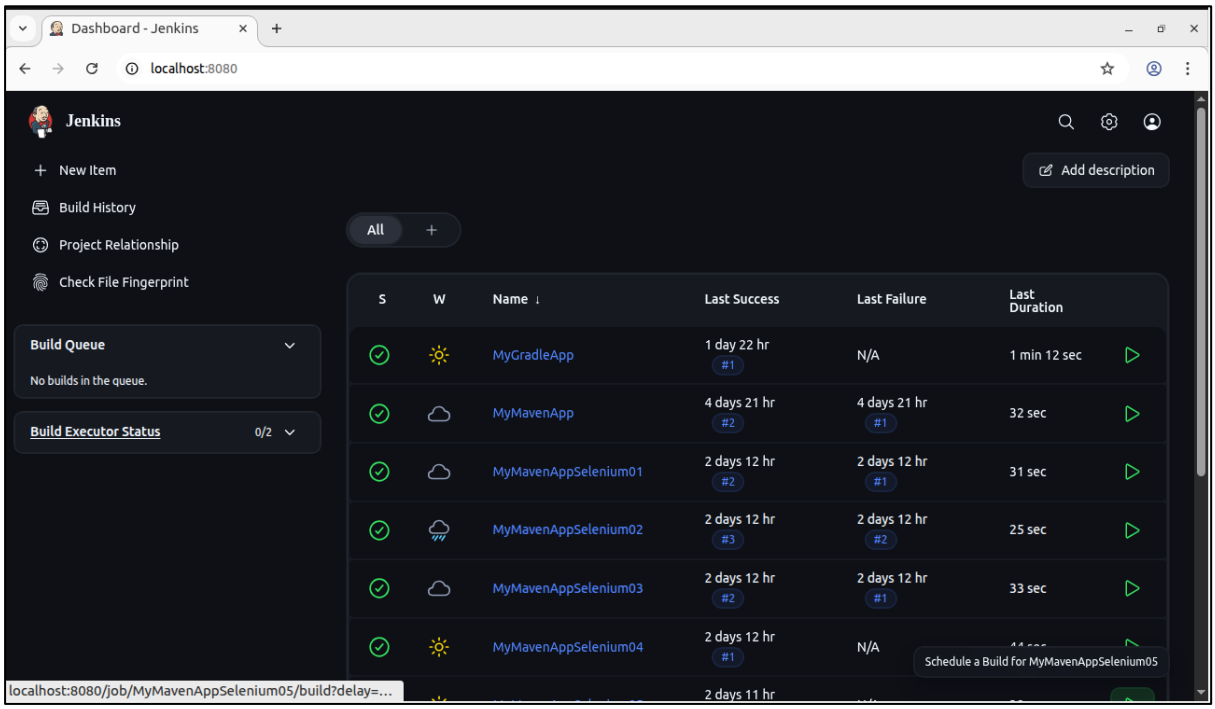
```
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ gradle run

> Task :run
Hello world.


BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
```

Step 6 : Execute gradle hello to run an specific task specified in the build.gradle file

```
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ gedit build.gradle
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ gradle hello

> Task :hello
good morning


BUILD SUCCESSFUL in 1s
1 actionable task: 1 executed
sumanth@sumanth-VirtualBox:~/Devops/MyGradleApp$ gradle run

> Task :run
Hello world.


BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
```

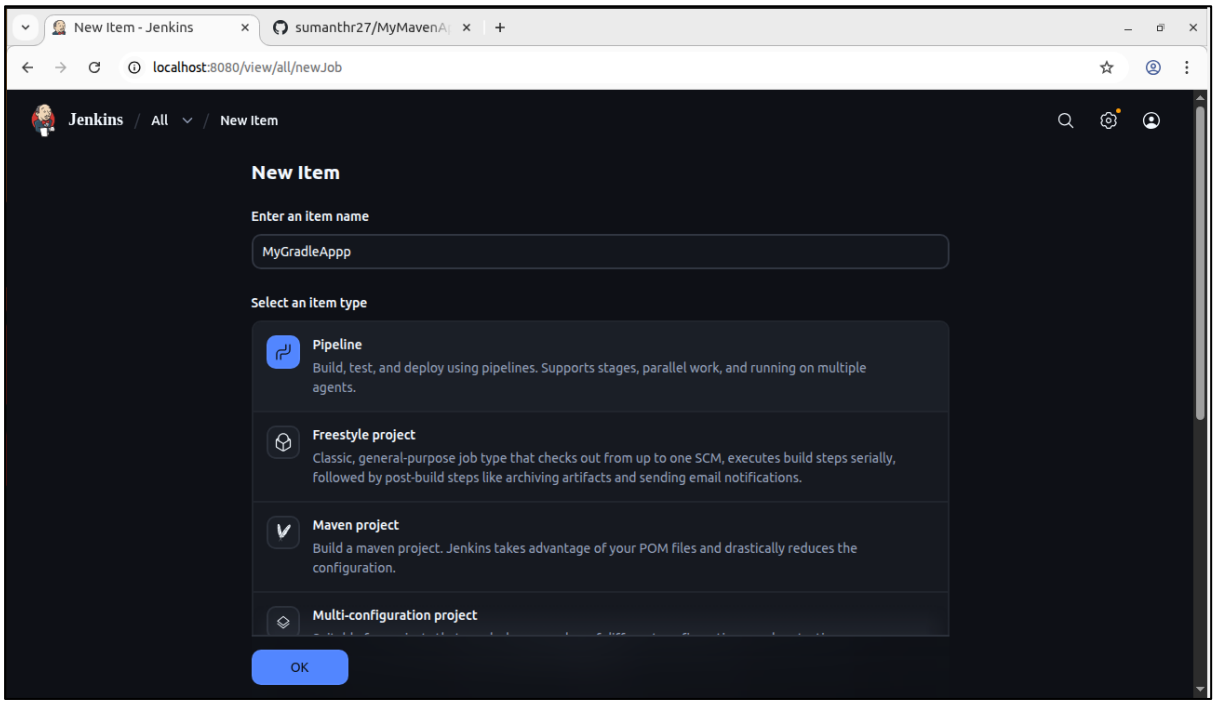
Jenkins Pipeline for MyGradleApp

Step 1: Open the Jenkins Dashboard in order to create a new item

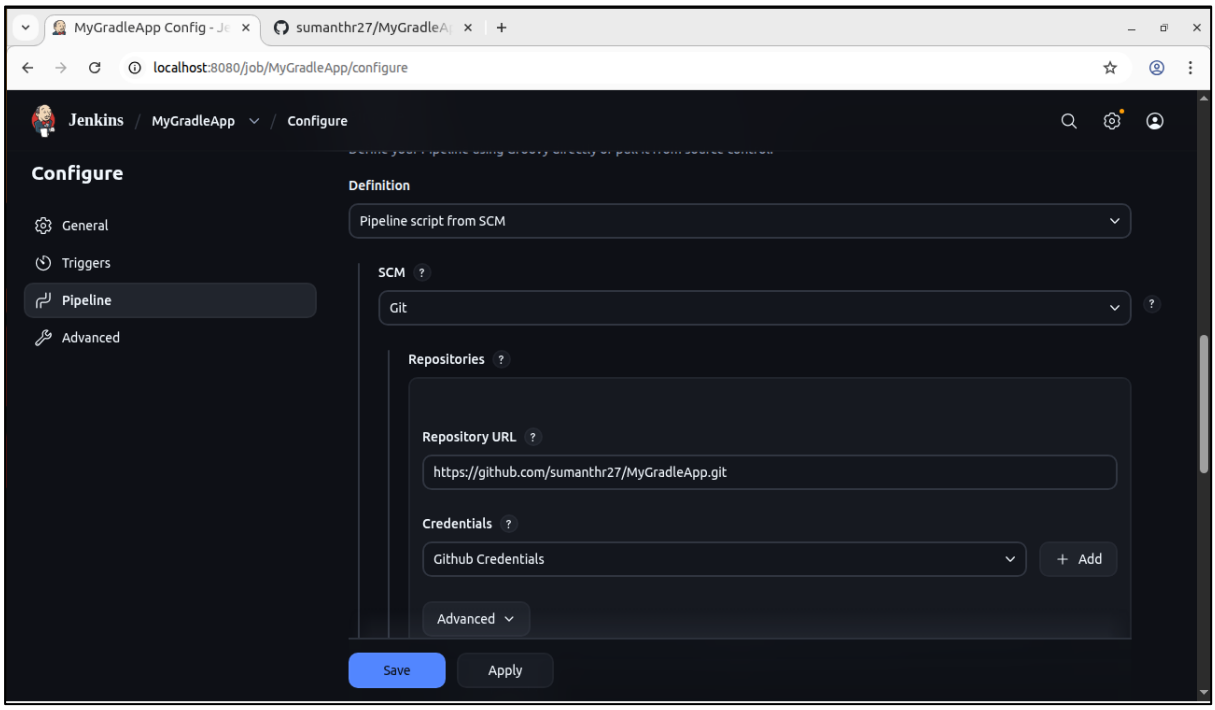


Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyGradleApp” and Item type as Pipeline and

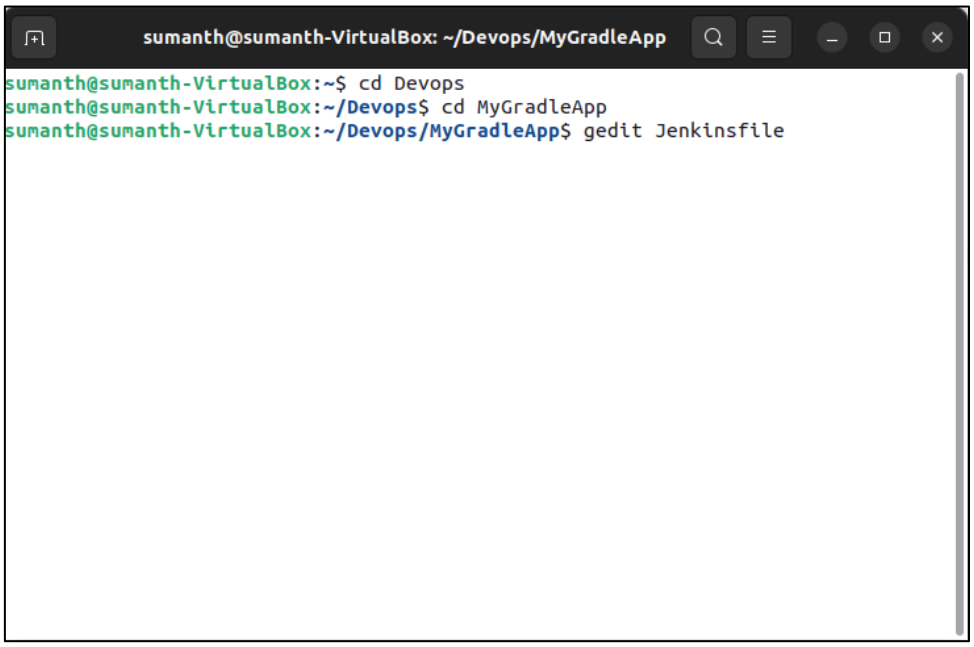
Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenAppSelenium05 and create an new file called “Jenkinsfile”



Step 5: Paste the available code inside the Jenkinsfile

Jenkinsfile
~/Devops/MyGradleApp

```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyGradleApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
```

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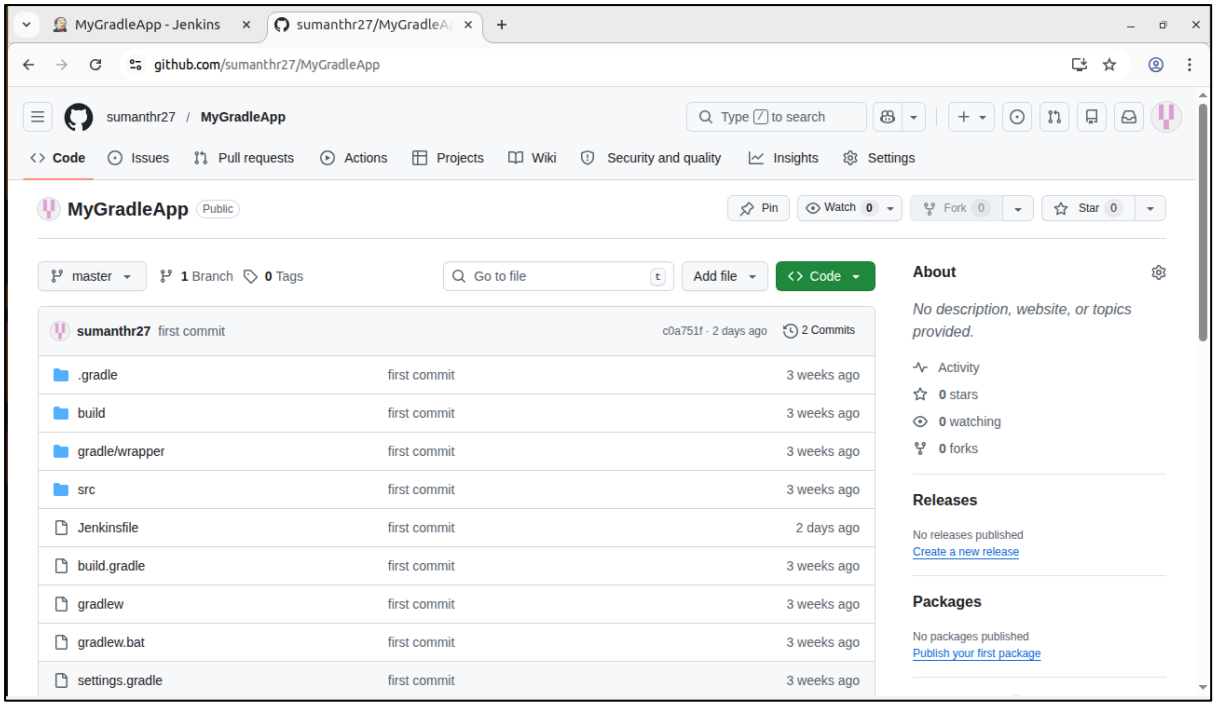
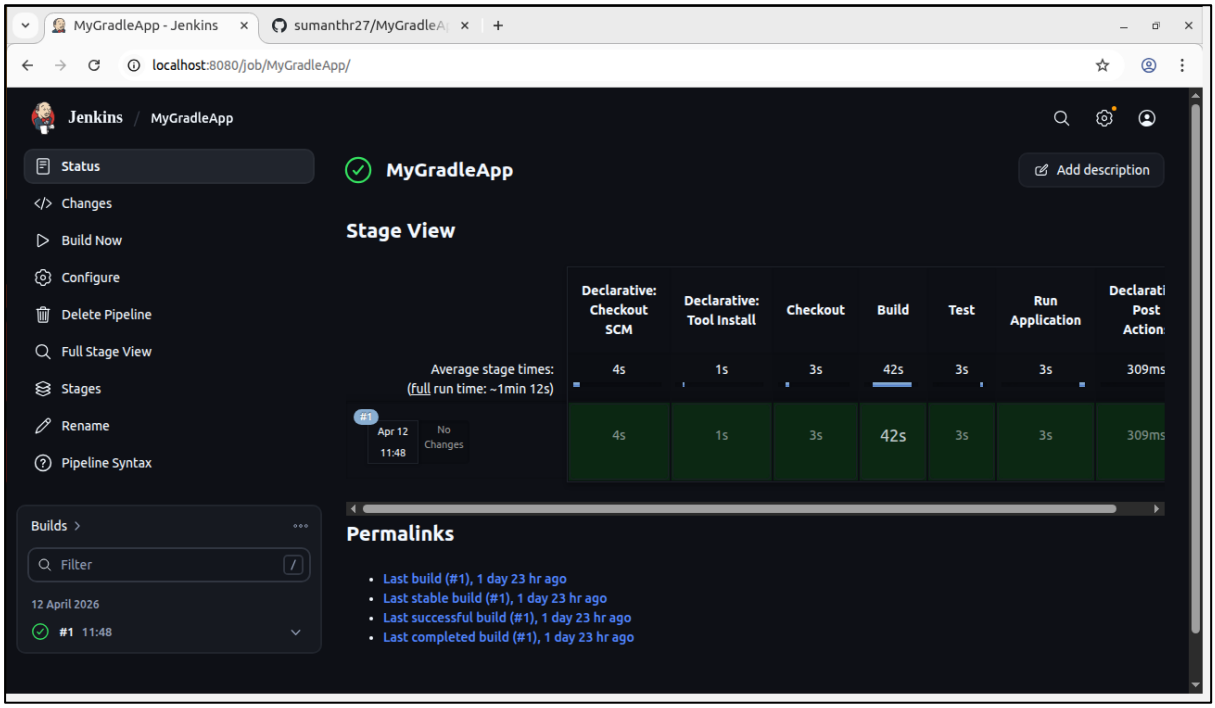
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.